

Imports

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```
library('astsa')
```

```
## Warning: package 'astsa' was built under R version 4.3.3
```

```
library('dplyr')
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
load("/Users/cream/Downloads/finalproject.Rdata")
```

```
head(finalPro_data)
```

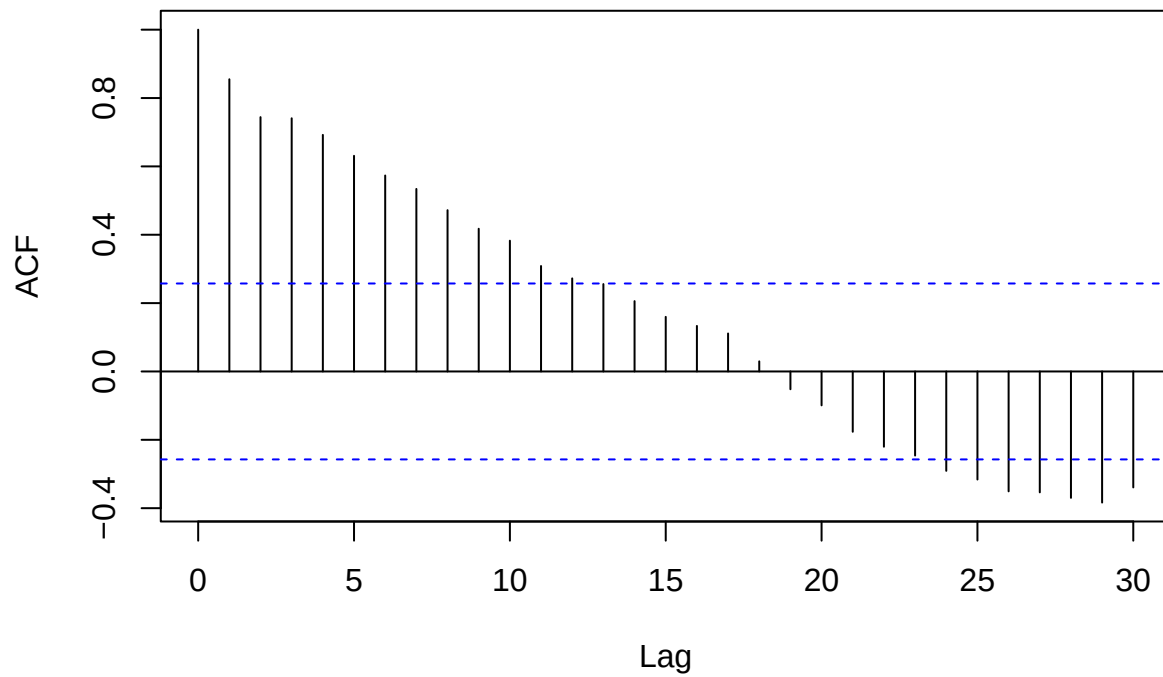
```
## # A tibble: 6 x 9
```

```
##   Country          Code  Year   GDP Growth  CPI Imports Exports Population
##   <fct>          <fct> <dbl>  <dbl>  <dbl> <dbl>  <dbl>  <dbl>
## 1 Central African Re~ CAF  1960 1.12e8 NA      NA    34.2   23.3   1503508
## 2 Central African Re~ CAF  1961 1.23e8 4.95    NA    35.8   26.5   1529227
## 3 Central African Re~ CAF  1962 1.24e8 -3.71   NA    37.7   24.6   1556661
## 4 Central African Re~ CAF  1963 1.29e8 -0.707  NA    38.5   25.2   1585763
## 5 Central African Re~ CAF  1964 1.42e8 2.08    NA    40.8   28.4   1616516
## 6 Central African Re~ CAF  1965 1.51e8 0.948   NA    37.7   27.1   1648833
```

```
imports = finalPro_data$Imports
```

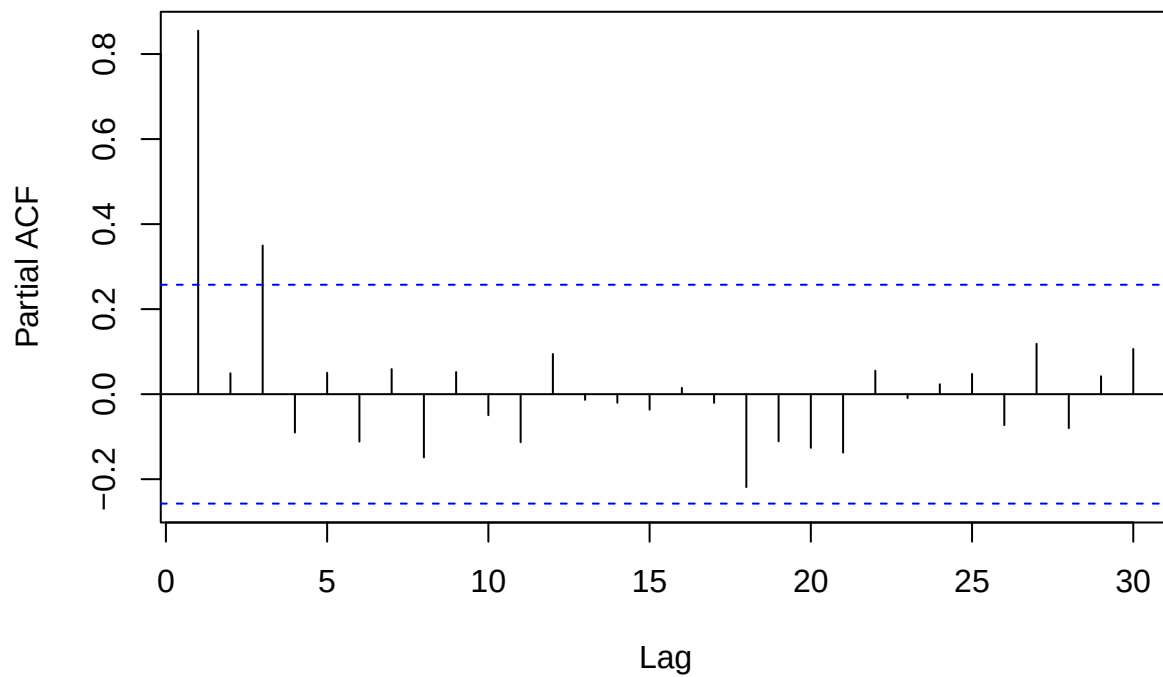
```
acf(imports, lag.max = 30, na.action = na.pass)
```

Series imports

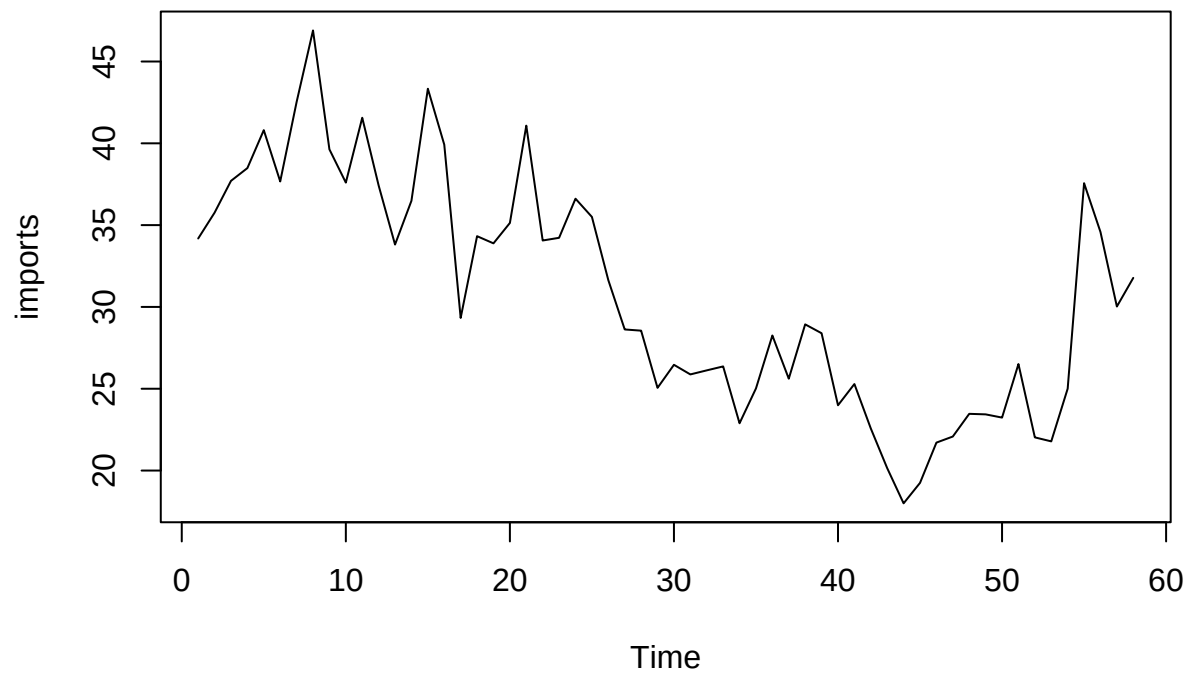


```
pacf(imports,lag.max = 30, na.action = na.pass)
```

Series imports



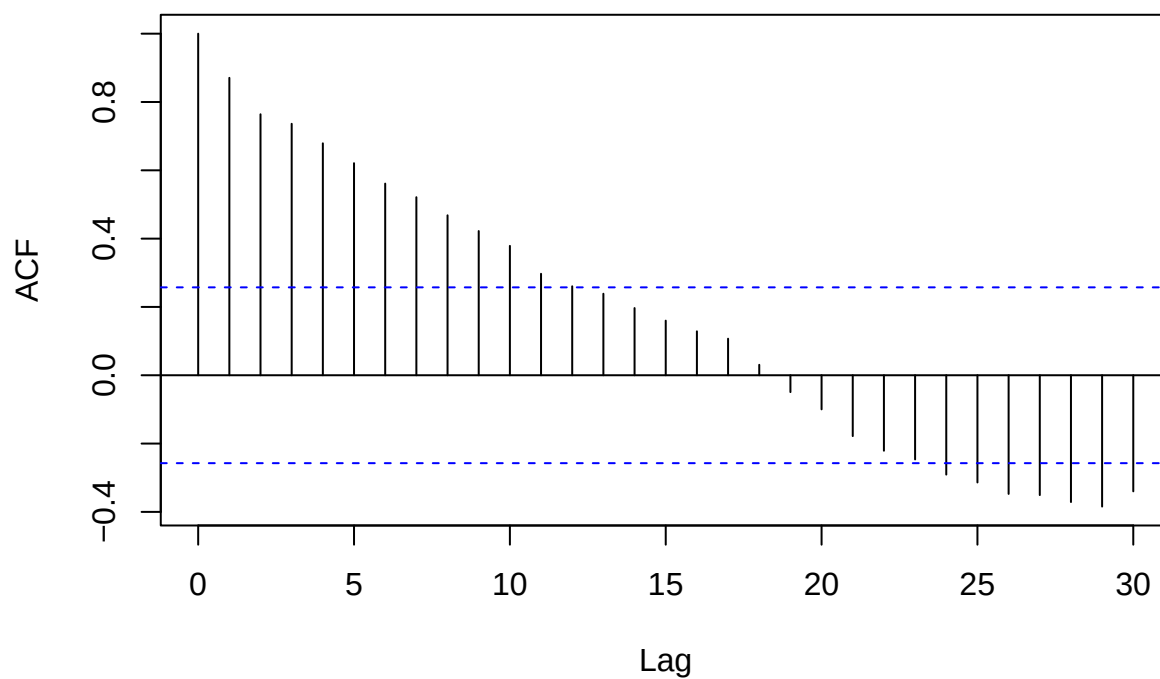
```
ts.plot(imports)
```



log imports

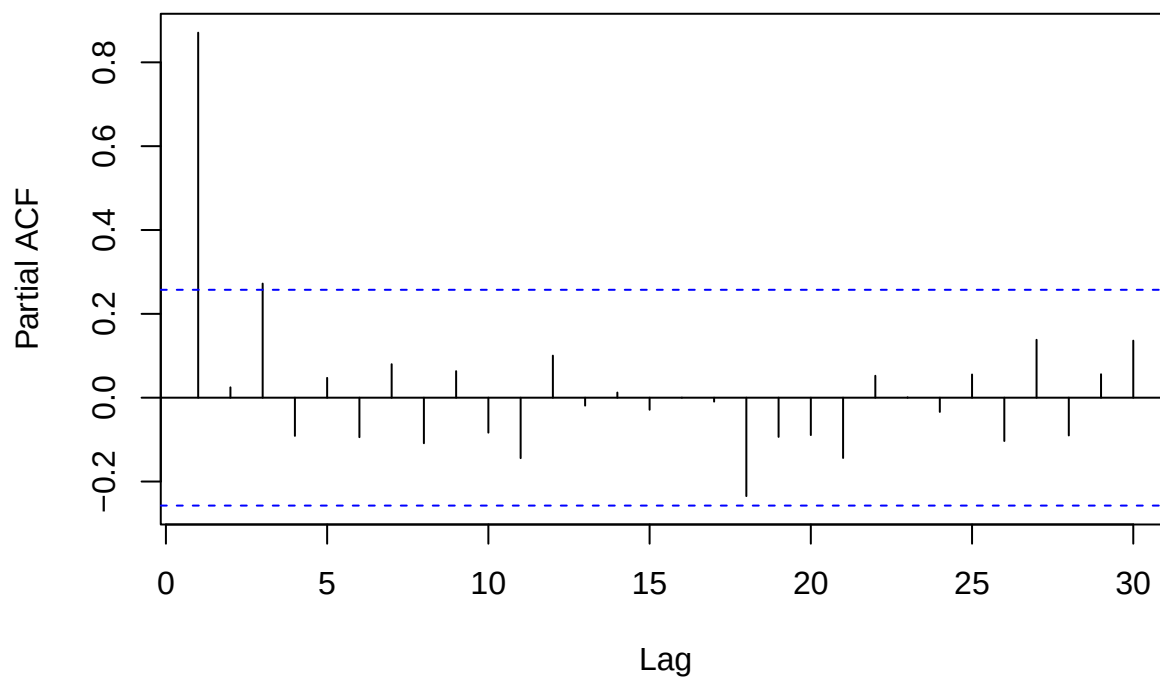
```
log.imports = log(imports)
acf(log.imports, lag.max = 30, na.action = na.pass)
```

Series log.imports

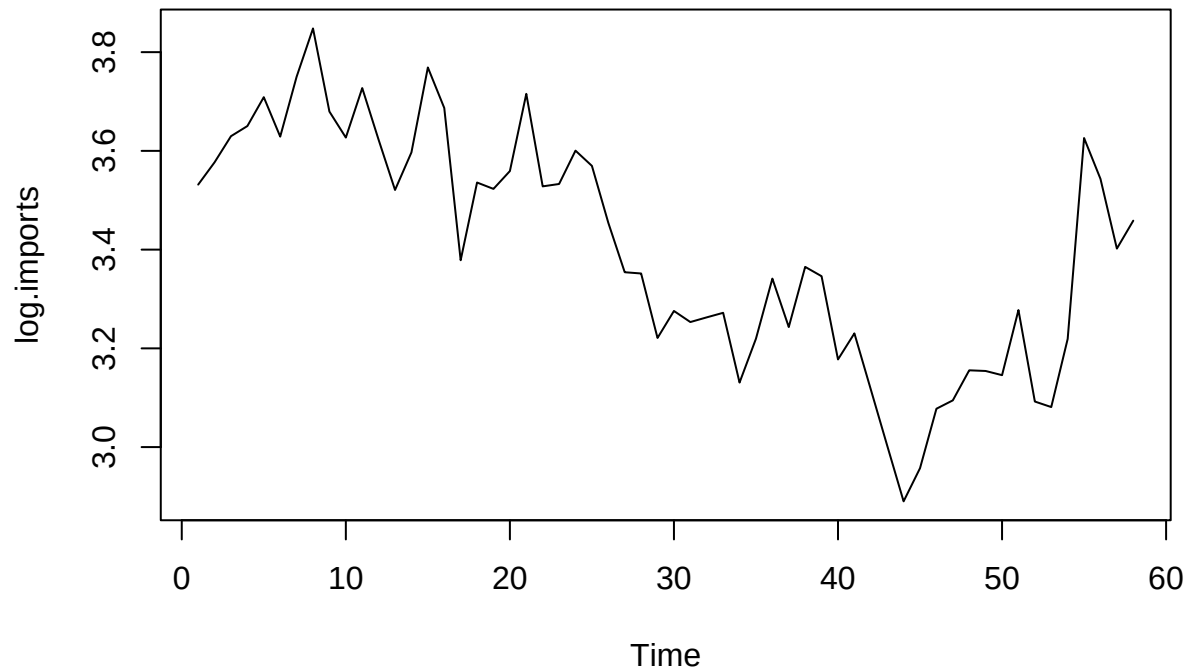


```
pacf(log.imports, lag.max = 30, na.action = na.pass)
```

Series log.imports

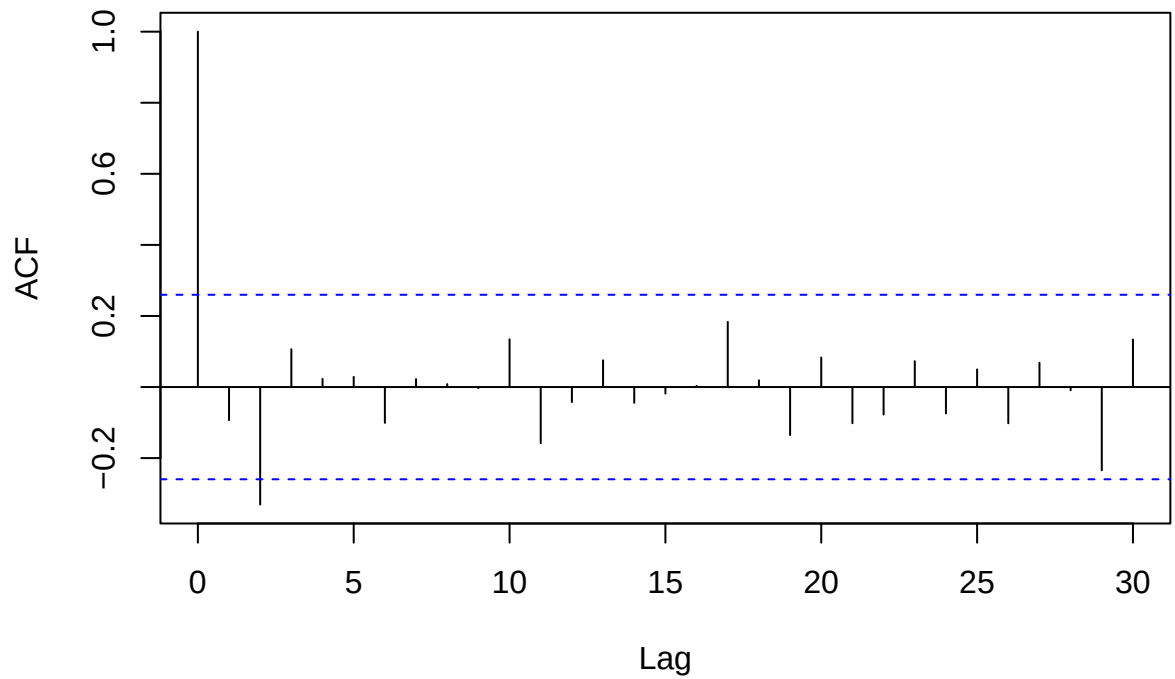


```
ts.plot(log.imports)
```



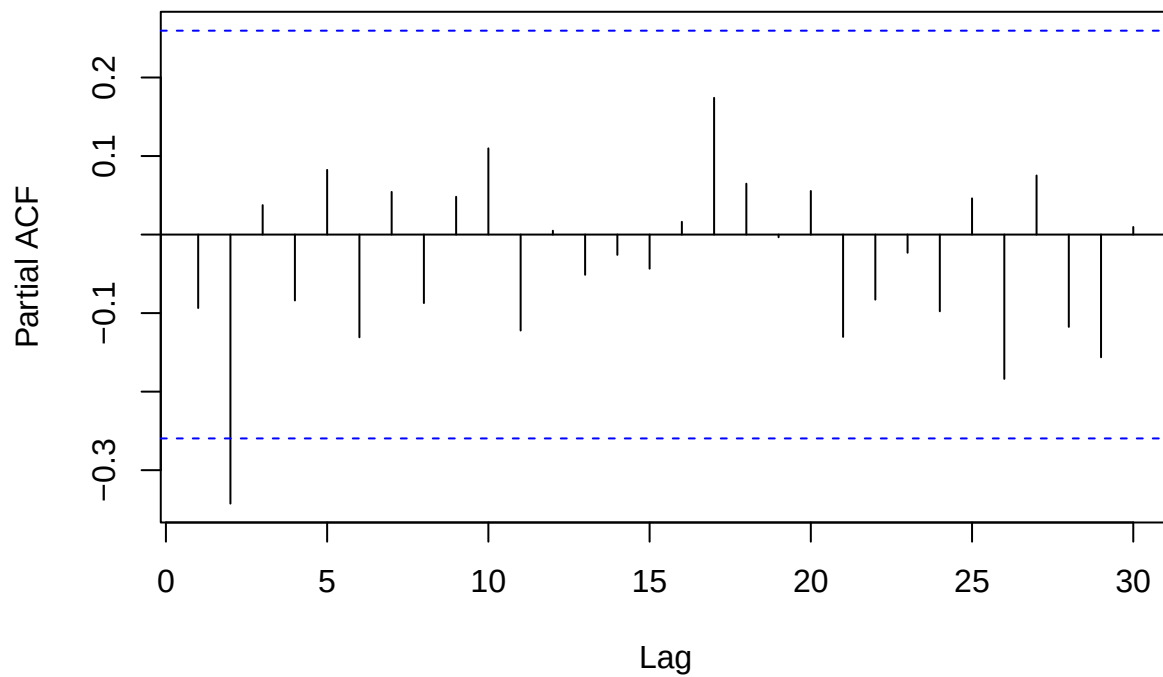
```
log.imports = ts(log.imports)
log.diff.imp = diff(log.imports, differences = 1)
acf(log.diff.imp, lag.max = 30, na.action = na.pass)
```

Series log.diff.imp

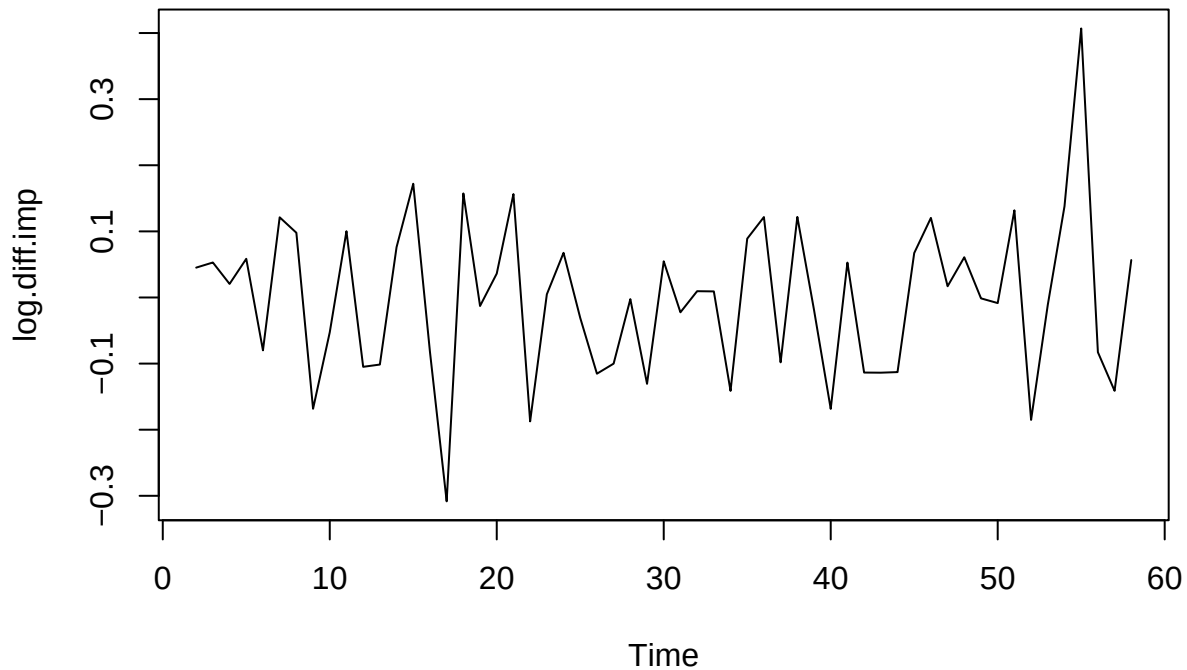


```
pacf(log.diff.imp, lag.max = 30, na.action = na.pass)
```

Series log.diff.imp



```
ts.plot(log.diff.imp)
```



- try:

`arma(1,1)` and `arma(1, 2)` and `arma(2,1)`

```
log.imp.arma21 = arima(log.imports, order = c(2, 1, 1))
sarima(log(finalPro_data$Imports), 2, 1, 1)
```

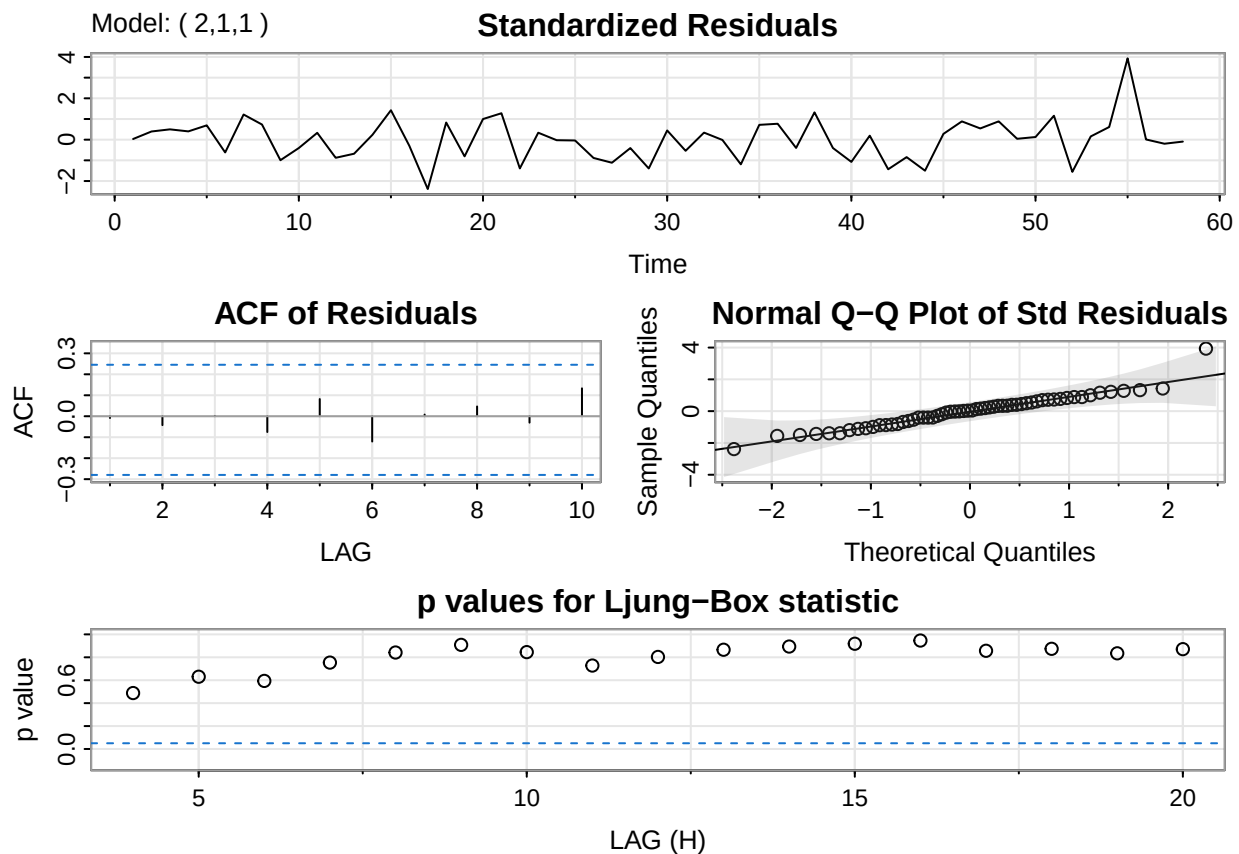
`arma(2,1)`

```
## initial value -2.117685
## iter 2 value -2.183588
## iter 3 value -2.185259
## iter 4 value -2.185930
## iter 5 value -2.186463
## iter 6 value -2.186556
## iter 7 value -2.187485
## iter 8 value -2.187673
## iter 9 value -2.187720
## iter 10 value -2.187726
## iter 11 value -2.187727
## iter 11 value -2.187727
## iter 11 value -2.187727
## final value -2.187727
## converged
## initial value -2.199962
## iter 2 value -2.200019
## iter 3 value -2.200062
## iter 4 value -2.200062
## iter 5 value -2.200063
```

```

## iter    6 value -2.200064
## iter    7 value -2.200069
## iter    8 value -2.200072
## iter    9 value -2.200074
## iter   10 value -2.200074
## iter   10 value -2.200074
## final   value -2.200074
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1         -0.3207  0.7160  -0.4479  0.6561
## ar2         -0.3531  0.1222  -2.8890  0.0056
## ma1          0.2253  0.8057   0.2797  0.7808
## constant    -0.0016  0.0108  -0.1486  0.8824
##
## sigma^2 estimated as 0.01221869 on 53 degrees of freedom
##
## AIC = -1.386832  AICc = -1.373337  BIC = -1.207617
##

```

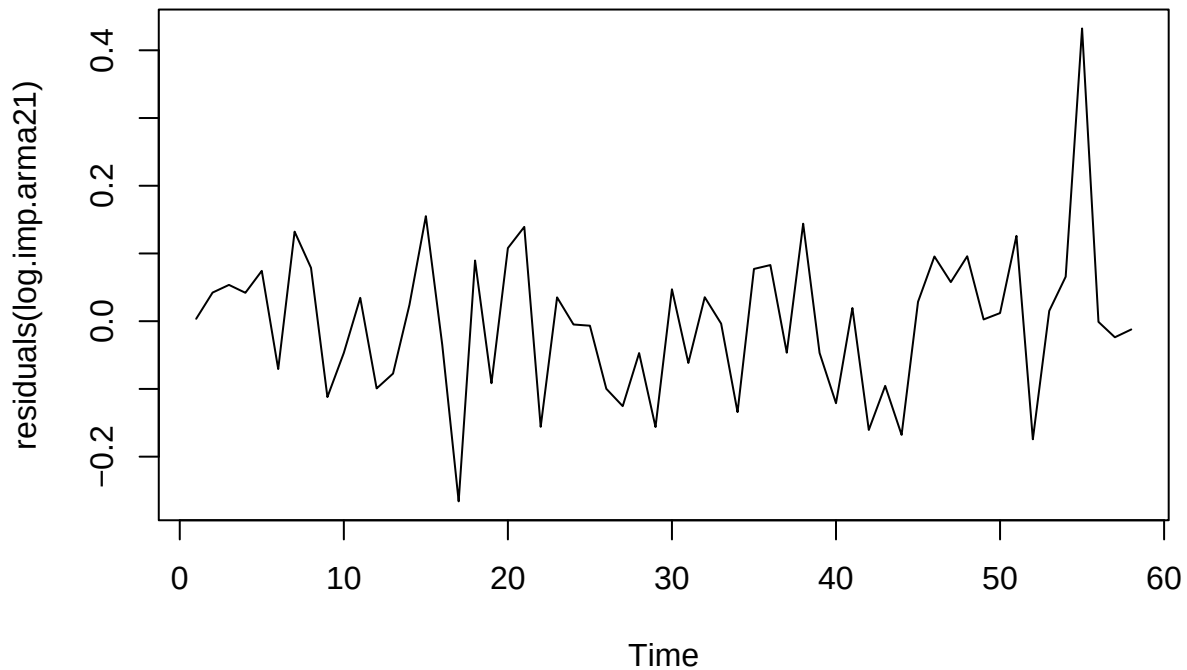


```
print(AIC(log.imp.arma21))
```

```
## [1] -81.02731
```



```
ts.plot(residuals(log.imp.arma21))
```



```
Box.test(residuals(log.imp.arma21), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
##  
## Box-Ljung test  
##  
## data: residuals(log.imp.arma21)  
## X-squared = 10.708, df = 20, p-value = 0.9534
```

```
log.imp.arma12 = arima(log.imports, order = c(1, 1, 2))  
sarima(log(finalPro_data$Imports), 1, 1, 2)
```

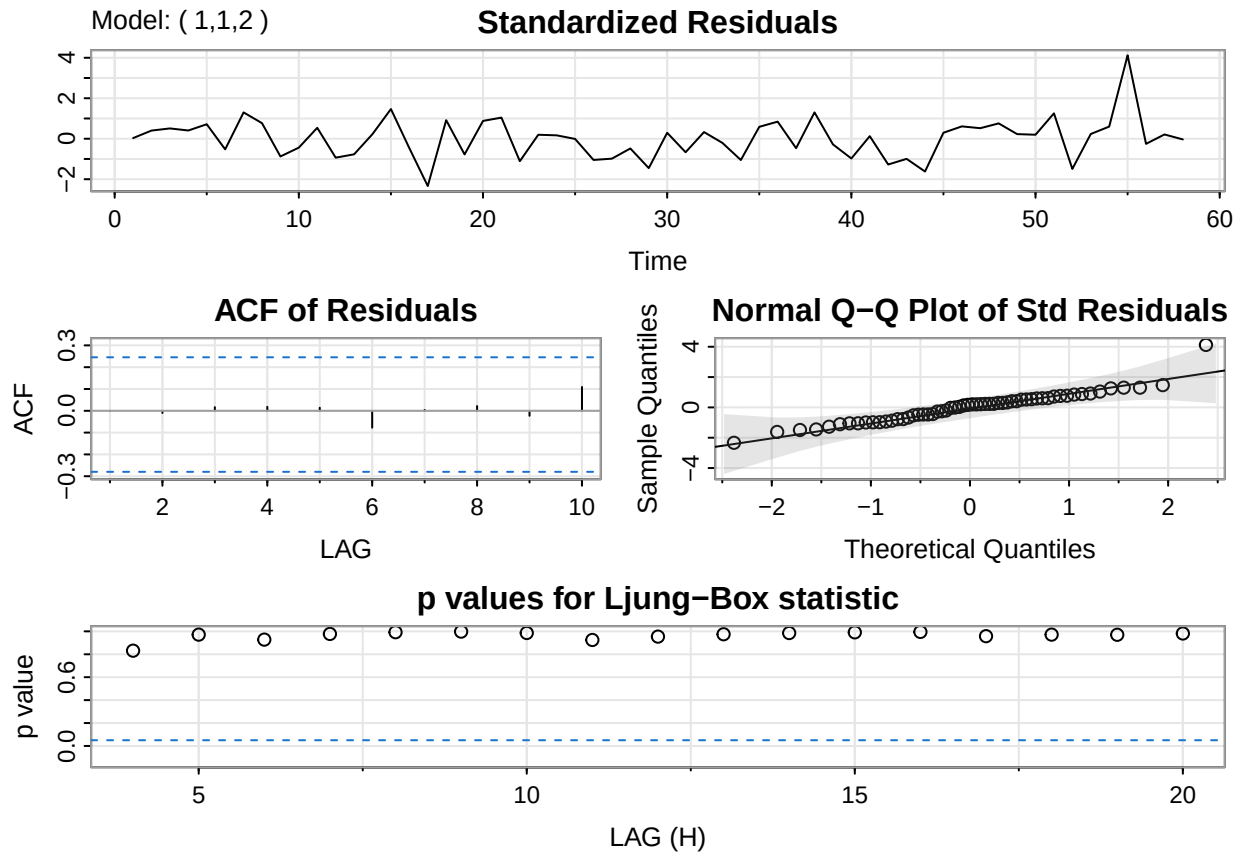
```
arma(1,2)
```

```
## initial value -2.124853  
## iter 2 value -2.190526  
## iter 3 value -2.196281  
## iter 4 value -2.197046  
## iter 5 value -2.198253  
## iter 6 value -2.199359  
## iter 7 value -2.201100  
## iter 8 value -2.202005  
## iter 9 value -2.202359  
## iter 10 value -2.202363  
## iter 11 value -2.202364  
## iter 12 value -2.202364
```

```

## iter 13 value -2.202364
## iter 14 value -2.202364
## iter 15 value -2.202364
## iter 15 value -2.202364
## iter 15 value -2.202364
## final value -2.202364
## converged
## initial value -2.209843
## iter 2 value -2.209920
## iter 3 value -2.210058
## iter 4 value -2.210177
## iter 5 value -2.210240
## iter 6 value -2.210268
## iter 7 value -2.210268
## iter 7 value -2.210268
## iter 7 value -2.210268
## final value -2.210268
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.3312 0.2902 -1.1412 0.2589
## ma1       0.2433 0.2720 0.8945 0.3751
## ma2      -0.4033 0.1267 -3.1820 0.0024
## constant -0.0025 0.0093 -0.2631 0.7935
##
## sigma^2 estimated as 0.01194653 on 53 degrees of freedom
##
## AIC = -1.407221 AICc = -1.393726 BIC = -1.228006
##

```



```
print(AIC(log.imp.arma12))
```

```
## [1] -82.14326
```

```
Box.test(residuals(log.imp.arma12 ), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
##
## Box-Ljung test
##
## data: residuals(log.imp.arma12)
## X-squared = 7.1817, df = 20, p-value = 0.996
```

```
log.imp.ma1 = arima(log.imports, order = c(0, 1, 1))
sarima(log(finalPro_data$Imports), 0, 1, 1)
```

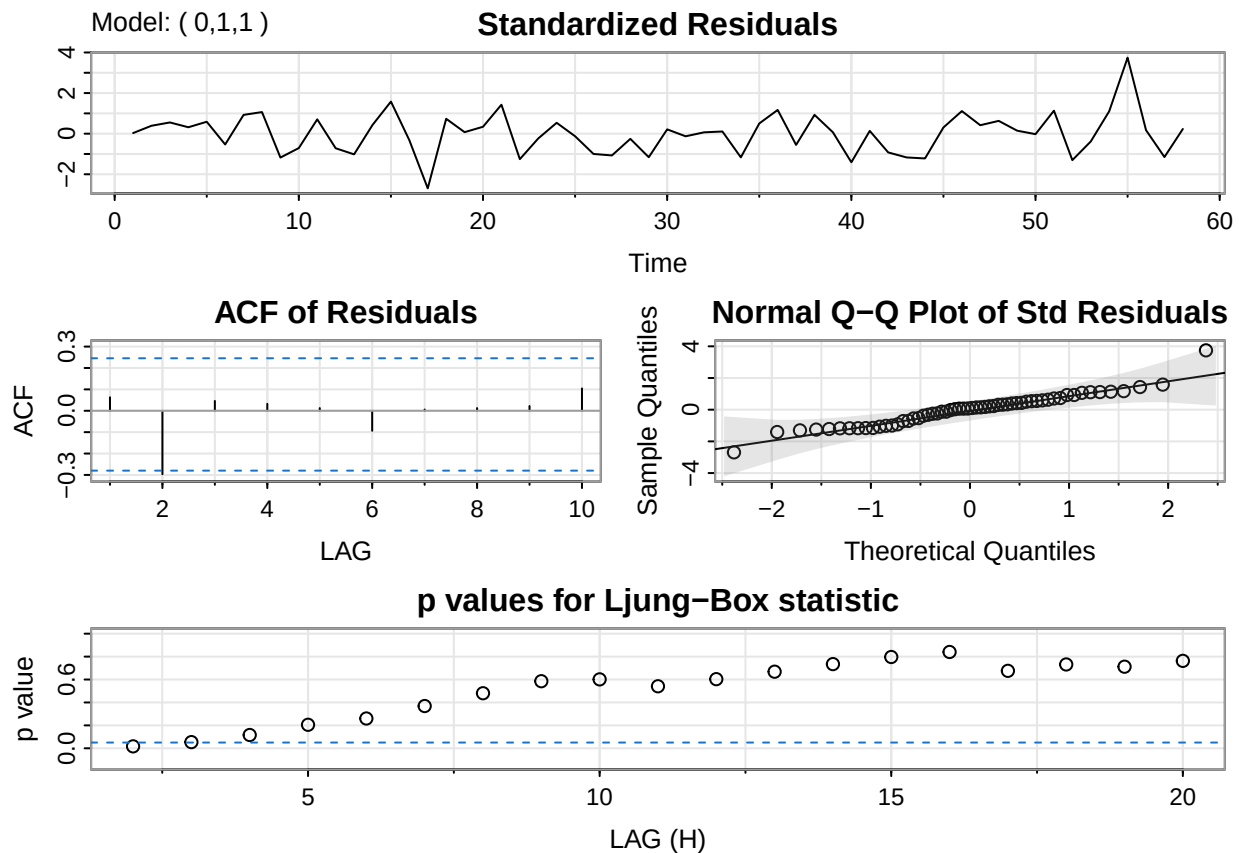
```
ma(1)
```

```
## initial value -2.132354
## iter 2 value -2.139659
## iter 3 value -2.143882
## iter 4 value -2.144278
## iter 5 value -2.144320
```

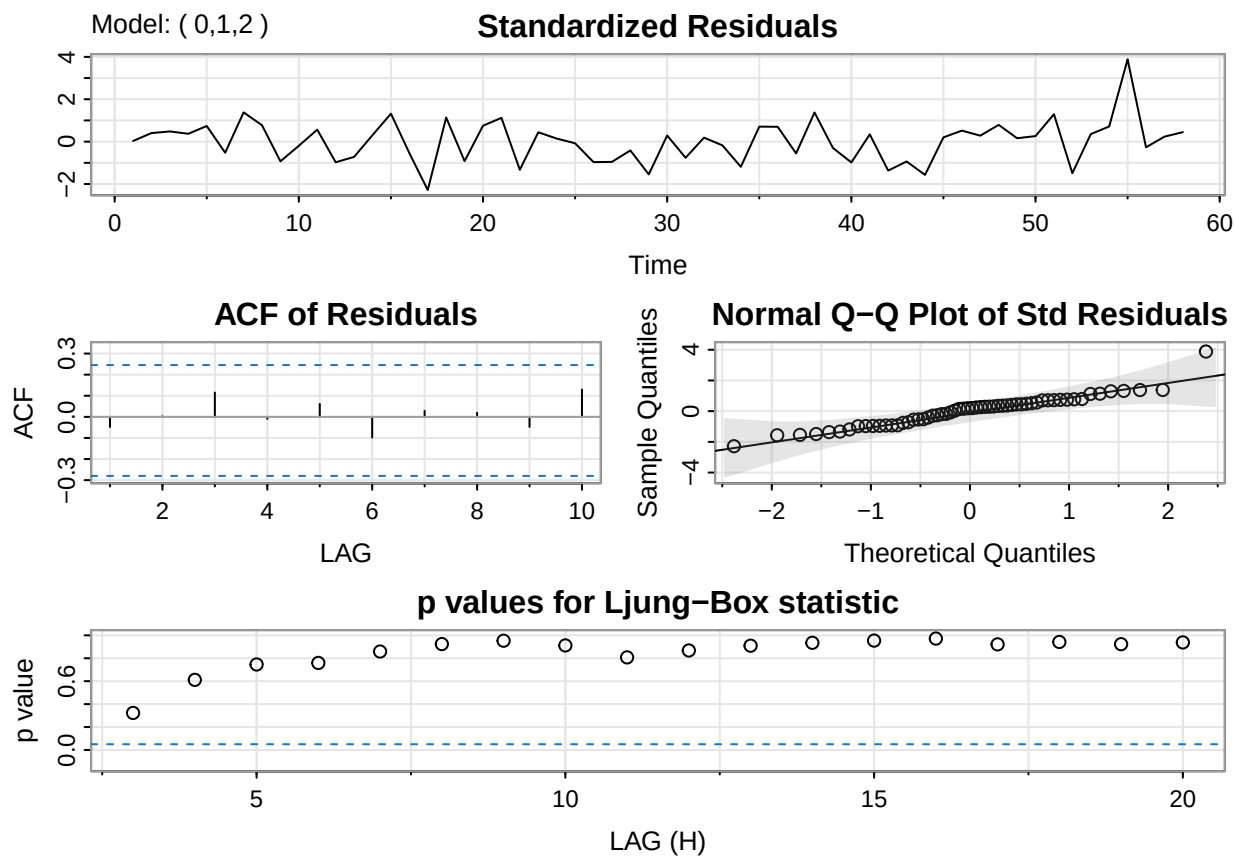
```

## iter 6 value -2.144360
## iter 7 value -2.144361
## iter 8 value -2.144361
## iter 9 value -2.144361
## iter 9 value -2.144361
## iter 9 value -2.144361
## final value -2.144361
## converged
## initial value -2.143999
## iter 2 value -2.144004
## iter 3 value -2.144005
## iter 4 value -2.144007
## iter 4 value -2.144007
## iter 4 value -2.144007
## final value -2.144007
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1         -0.2314 0.1836 -1.2604 0.2128
## constant    -0.0016 0.0120 -0.1367 0.8917
##
## sigma^2 estimated as 0.01371893 on 55 degrees of freedom
##
## AIC = -1.344873  AICc = -1.340975  BIC = -1.237344
##

```



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```
AIC(log.imp.ma2)
```

```
## [1] -83.02682
```

```
Box.test(residuals(log.imp.ma2), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
##
## Box-Ljung test
##
## data: residuals(log.imp.ma2)
## X-squared = 9.8844, df = 20, p-value = 0.9702
```

```
arma(1,1,0): -78 arma(0,1,2) : -83.02682 arma(2, 1, 2): -80.15922
```

Forecast

```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.3.3
```

```
## Registered S3 method overwritten by 'quantmod':
## method from
## as.zoo.data.frame zoo
```

```
##
## Attaching package: 'forecast'

## The following object is masked from 'package:astsa':
##
##     gas
```

```
forecast(log.imp.ma2,level=c(80,95))
```

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 59	3.450150	3.308524	3.591776	3.233552	3.666748
## 60	3.433593	3.237578	3.629609	3.133813	3.733373
## 61	3.433593	3.221180	3.646007	3.108735	3.758451
## 62	3.433593	3.205960	3.661226	3.085459	3.781728
## 63	3.433593	3.191696	3.675490	3.063644	3.803543
## 64	3.433593	3.178228	3.688958	3.043046	3.824141
## 65	3.433593	3.165435	3.701751	3.023481	3.843706
## 66	3.433593	3.153225	3.713961	3.004808	3.862379
## 67	3.433593	3.141526	3.725661	2.986915	3.880272
## 68	3.433593	3.130277	3.736909	2.969711	3.897475

```
autoplot(forecast(log.imp.ma2))
```

